Electronic Principles and Devices · Unit_01 · Unit-1 Semiconductor Diode

DRAFT — UE26EC141A · Unit_01 · Unit-1 Semiconductor Diode · Logical Operations using diode

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Logical Operations using diode

1 Logical Operations using diode

Study focus: Logical Operations using diode. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000062 · EV0000063

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

I'm Dr.

Key details

- Anand, Department of Electronics and Communication.
- Today, we're going to see the logical operations.
- A logical operation is an operation that acts on a binary number to produce the result according to the laws of Boolean logic.
- A logic gate is a basic building block of digital circuits.
- Logic gate is considered as a device that has the ability to produce one output level with a combination of input levels.
- The function is represented f and Y is the result.
- In continuation, logic gates are implemented by using diode transistors and by other several devices.
- This logic gate can also be considered as electronic circuits.
- The logic gate opened by diode is called diode resistive logic.
- Inputs and outputs of logic gates are in two levels, which are termed as high and low, or true or false, or ON and OFF, simply one and zero.
- Coming to the next step.
- For all logic gates, a table with all the combination is listed below.
- The combination of input variable and the corresponding output is termed as truth table, which is shown here.
- It explains how the logic circuit output responds to various combination of logic level at the input.
- AND operation is explained by the following truth table, where.

- AND operation is A.B.
- When inputs A and B are zero, zero, B results in zero.
- Input zero and one multiplied to get zero.
- One and zero, zero.
- One and one is one.
- This is also explained by a Venn diagram here.
- The common is being highlighted here with the red color.
- OR gate produces an output of logic one as shown in the diagram.
- Even if one of the input is one state and produces an output of zero state if all the inputs are in the logic zero state.
- OR operation is explained with the following table.
- With zero, zero, we get zero.
- With zero and one, and one and zero, and one and one, we get all one states.
- The two diodes are connected in parallel here, and they're connected in series with a resistance with 1K.
- With the input E is equal to 10, with input.
- E1 and E2 here.
- And that we can represent using a truth table with the input V1 in the one column,.
- V2 second column, and V output in the third column.
- The input V1 is also called as an input A, and similarly input two respectively with respect to V1 and V2.
- When both the inputs are zero, zero, diodes are not forward biased and hence Vout is equal to zero.
- If one of the input is 10 volts, so as shown in the diagram here, the first V input is 10 volts here, the second is zero, the diode one is forward biased hence we represent 0.7.
- And what the difference between the applied voltage here and we calculate the Vout.
- It's not 0.3, it's 0.7.
- Here diodes are reverse connected compared to the previous one, and we connect a voltage E is equal to 10 volts here.
- And applied voltage E1 and E2.
- Because the difference between here is 0.7, so we get 0.7.

Example or application from the lecture

- So we can build an example of AND, OR NOT functions.
- Now consider the first condition here, zero, zero.
- If one of the voltage we consider 10 volts here, so we consider this 10 volts and the second voltage is zero, diode is on.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000062 · EV0000063 · EV0000064 · +1 source units in pack.json

Core explanation · 05:00–09:59

The previous case also diode was on, so now second diode is on, and hence the difference in the voltage is 0.7 we get.

Key details

- If both the diodes are applied with E1 and E2, 10 volts, then the both diodes are off, and hence the voltage across this is 10 volts.
- AND with the help of a diode.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000066 · EV0000067 · EV0000068 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Logical Operations using diode transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9977; only the JSON segmentation supplies the displayed timestamps.
Logical Operations using diode transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26EC141A_Electronic_Principles_and_Devices/Unit_01/AV_Summaries/008_Logical_Operations_using_diode__a5ae846222c4060c.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
2. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
3. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
4. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00030	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare...
SRC00031	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00032	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare...
SRC00033	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0065. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0066 to s0075. Sidecar variants compare...

C Revision and retrieval practice